



LV.0 → LV.1 · AI ENGINEER

7 V-words that move you from **Level 0** → 1.

Vectors, VAE, Vector Embeddings, Vision Transformer & more.

Plain-English definitions, one swipe each, plus the V-named tools
and minds to follow.

↓ UP NEXT • TERMS

The V stack.

Vectors

VAE

Vector Embeddings

Vision Transformer

Vs of Big Data

Vectorization

Vector Database

V

VS OF BIG DATA • FOUNDATIONS

Vs of Big Data

Volume • Velocity • Variety • Veracity • Value

A framework describing the key characteristics of big data systems, **Volume, Velocity, Variety, Veracity, and Value**, which together define the scale, speed, diversity, reliability, and usefulness of

- Data engineering
- Analytics platforms
- AI infrastructure



Drop diagram • Vs of Big Data

5 VS FRAMEWORK

VARIATIONAL AUTOENCODER • GENERATIVE AI

VAE

compress • capture distribution • generate

A generative deep learning model that learns compressed representations of data while capturing underlying probability distributions, able to **generate new samples similar to its training data**. Used in image generation, anomaly detection, and

- Image generation
- Anomaly detection
- Representation learning



Drop diagram • VAE

ENCODE → LATENT → DECODE

Vectors

data as numbers in space

Mathematical representations of data as ordered numerical values in multidimensional space. In AI, vectors represent text, images, and audio, **enabling similarity search, machine learning, and semantic understanding.**

• Machine learning

• Similarity search

• Recommendations



Drop diagram • Vectors

POINT IN N-DIMENSIONAL SPACE

Vectorization

convert raw data into vectors

The process of converting raw data such as text, images, or audio into **numerical vector representations** in multidimensional space, so models can compare, cluster, and retrieve it.

USED IN

- NLP pipelines
- Feature engineering
- Search



Drop diagram • Vectorization

TEXT / IMAGE → [0.2, 0.8, ...]

VECTOR DATABASE • INFRASTRUCTURE

Vector Database

store embeddings • search by similarity

A database built to store and query vector embeddings, retrieving results by **similarity rather than exact match**. The backbone of semantic search and retrieval-augmented generation.

USED IN

- Semantic search
- RAG
- Recommendations



Drop diagram • Vector Database

QUERY VECTOR → NEAREST NEIGHBORS

VECTOR EMBEDDINGS • EMBEDDINGS

Vector Embeddings

semantic meaning as numbers

Numerical representations of text, images, audio, or other data in a high-dimensional vector space. Embeddings **capture semantic meaning and relationships** between data points, making them essential for search, retrieval, clustering, and generative AI.

USED IN

- Semantic search
- RAG
- Clustering



Drop diagram • Vector Embeddings

SIMILAR MEANING • CLOSE VECTORS

VISION TRANSFORMER • ARCHITECTURE

Vision Transformer

image patches • self-attention • no convolutions

A deep learning architecture that adapts the Transformer for computer vision. Instead of convolutions, it **divides images into patches and processes them with self-attention**, enabling strong results in classification, detection, segmentation, and multimodal AI.

USED IN

- Image classification
- Detection
- Multimodal AI



Drop diagram • Vision Transformer

IMAGE PATCHES → TOKENS → ATTENTION

↓ UP NEXT • TOOLS & COMPANIES

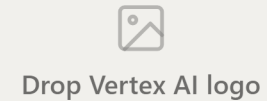
The V stack.

Three V-named tools worth knowing, from **Google's unified ML platform** to the embedding models powering retrieval, plus a place to learn AI itself.



ML PLATFORM • GOOGLE CLOUD

Vertex AI



Google Cloud • build • train • deploy

Google Cloud's unified ML and AI platform for **building, training, deploying, and managing models**, with tools for generative AI, MLOps, model serving, and enterprise-scale development.

Google

Unified GCP ML platform

GenAI

Generative AI and model serving

MLOps

Data prep, training, deployment

WHAT IT IS

End-to-end ML platform on Google Cloud.

IN AI STACKS

MLOps, fine-tuning, and model serving.

WHY TEAMS USE IT

One place to train, tune, and deploy.

FIND THEM

cloud.google.com/vertex-ai

EMBEDDING MODELS

Voyage AI

state-of-the-art embeddings • RAG



Drop Voyage AI logo

An AI company building **state-of-the-art embedding and retrieval models** optimized for search, RAG, and enterprise AI, known for high-performance embeddings that sharpen semantic search and knowledge retrieval.

Embeddings

SOTA embedding models

RAG

Built for retrieval quality

Enterprise

Search and knowledge retrieval

WHAT IT IS

High-performance embedding and retrieval models.

IN AI STACKS

The embedding layer feeding retrieval.

WHY DEVS USE IT

Better embeddings, better RAG results.

FIND THEM

voyageai.com

AI EDUCATION PLATFORM

Vizuara

learn AI • projects • mentorship



Drop Vizuara logo

An AI-focused education and research platform helping students and professionals **learn AI, ML, data science, and software engineering** through mentorship, projects, internships, and industry-oriented training.

Learn

AI, ML, and data science

Projects

Hands-on, industry-oriented

Mentor

Guidance and internships

WHAT IT IS

AI education and research platform.

BEST FOR

Going from beginner to job-ready.

WHY LEARNERS USE IT

Structured, project-based AI training.

FIND THEM

vizuara.ai

↓ UP NEXT • PEOPLE TO FOLLOW

The V minds.

Three V-named figures worth following, from a
founder of statistical learning theory to leaders in
responsible AI and generative AI engineering.



FOUNDER OF STATISTICAL LEARNING

Vladimir Vapnik

SVMs • Statistical Learning Theory



Vladimir Vapnik photo

A founding figure of modern machine learning who co-developed **Statistical Learning Theory and Support Vector Machines**, establishing mathematical foundations that still shape ML and pattern recognition today.

SVM

Co-invented Support Vector Machines

SLT

Statistical Learning Theory

Founder

Of modern ML theory

KNOWN FOR

SVMs and statistical learning theory.

BEST FOR

Foundational ML theory.

WHY FOLLOW

The mathematical roots of ML.

FIND THEM

[scholar.google.com / Vladimir Vapnik](https://scholar.google.com/VladimirVapnik)

RESPONSIBLE AI LEADER

Virginia Dignum

Umeå University • responsible AI



Virginia Dignum photo

An AI researcher, professor, and global leader in responsible AI whose work on **ethics, governance, and trustworthy AI systems** helps shape international discussion on safe AI deployment.

Ethics

Responsible and trustworthy AI

Governance

Shapes global AI policy

Professor

Human-centered AI research

KNOWN FOR

Responsible AI and AI governance.

BEST FOR

AI ethics, safety, and policy.

WHY FOLLOW

The ethics layer of every AI system.

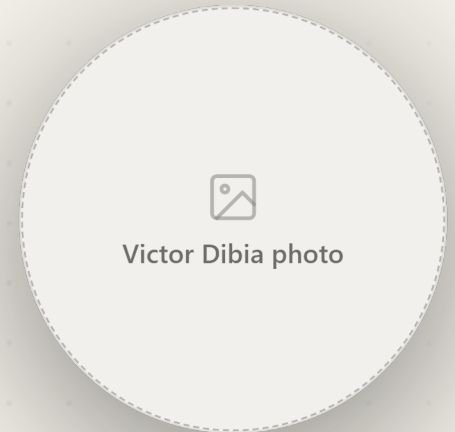
FIND THEM

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GENERATIVE AI ENGINEER

Victor Dibia

researcher • author • educator



An AI researcher, engineer, author, and educator working in **generative AI, LLMs, AI engineering, and developer tooling**, helping developers understand and build practical AI-powered applications.

GenAI

LLMs and AI engineering

Author

Writes on building with AI

Tooling

Developer-focused AI tools

KNOWN FOR

Practical generative AI engineering.

BEST FOR

Building real AI applications.

WHY FOLLOW

Hands-on LLM and tooling insight.

FIND THEM

linkedin.com/in/dibia

The **V** stack, in one card.

Save it. Drop a comment with the V-term you use most.

01 **Vs of Big Data** foundations

02 **VAE** generative

03 **Vectors** foundations

04 **Vectorization** data

05 **Vector Database** infra

06 **Vector Embeddings** embeddings

07 **Vision Transformer** architecture

+ **Vertex AI • Voyage AI • Vizuara** tools

@ **Vladimir Vapnik • Virginia Dignum • Victor Dibia** people



FOLLOW FOR THE FULL A TO Z
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W Next